SE

REFRACTIVE SURGERY

StandardEyes
Refractive surgery analysis

User manual & walkthrough · v1.1.0 · standardeyes.github.io

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What this app does

One tool from raw Excel to publication-ready tables and figures

1 **Compare groups**

Summary tables for any number of groups, with the test chosen per variable type and clustered by patient when both eyes of a patient are analyzed. P values in journal style (.27, <.001).

2 **Match fairly**

Matching with MatchIt (nearest neighbor on the logit of the propensity score, exact, coarsened exact or subclassification), with a live matched-count preview for nearest neighbor and balance reporting.

3 **Model properly**

GEE clustered on the patient, with the Mancl-DeRouen small-sample correction, so both eyes of a patient are handled correctly, plus covariate-adjusted regression.

4 **Publish**

Journal-style figures with every group on one chart, JCRS 2025 double-angle astigmatism plots (preoperative, SIA, prediction error), the astigmatism sheets, and stability over time.

Before you start

Almost everything is already inside the folder

You need R 4.5.x

The statistics run in R, so R must be installed on your computer.

Use R 4.5.x, not 4.6 or an older R. CRAN's main page now offers 4.6. R 4.5.2 is at cran.r-project.org/bin/windows/base/old/4.5.2/

Typical location:

C:\Program Files\R\R-4.5.2\

The app finds it automatically. If it can't, it asks you — see the next slide.

Already included

- The Python runtime that serves the app
- All 48 R packages the analysis needs
- The full statistical engine and figure code

Nothing is installed on your machine and nothing is downloaded — it works on a restricted network.

Starting the app

Copy once, then start it the same way every time

1

Copy the whole folder

Download StandardEyes-windows.zip from standardeyes.github.io, right-click it, choose Properties and tick Unblock. Extract the whole folder where you can write, e.g. C:\StandardEyes or Documents, not C:\Program Files.

2

Double-click the launcher

Start StandardEyes.bat

3

Work in your browser

It opens automatically on a free port, shown in the console window — so it never clashes with another app.

Keep the black console window open while you work. Closing it stops the app.

If the app can't find R

A one-time Setup screen appears — you can also reopen it any time

You have three ways to point it at R:

- **Pick from the list** — Any R installation found on the computer is listed — click one.
- **Browse…** — Opens a file picker. Choose Rscript.exe inside R's bin folder.
- **Paste the path** — Type it in and press Use this.

The app checks the choice immediately and reports the R version it found, with a warning if it is not 4.5.x.

No R at all? Install R 4.5.2 for Windows from cran.r-project.org/bin/windows/base/old/4.5.2/ (the main CRAN page offers a newer R), then click Re-check. No extra R packages are ever needed — this app brings its own.

Reopen later with the R setup button at the top right.

The workflow

Eight tabs, left to right — top to bottom here

1

Data & Groups

Load Excel, define groups

2

Variables & Types

Sections, types, VA notation

3

Filters & Subsets

Inclusion rules + subsets

4

Matching

Propensity-score matching

5

Regression

Adjusted GEE

6

GEE

Clustered models, table tests, P style

7

Special outputs

Figures, astigmatism, column roles

8

Run & Results

Check, run, preview, save

Steps 1–2 · Data and variables

1 Data & Groups

Load your Excel file (.xlsx or .xlsm) with Browse & load Excel in the left sidebar, then pick the sheet. The file stays on your computer.

Choose the column that defines the groups, then map its values into the groups you want to compare (in the example: Reference, Cohort A, Cohort B).

Set the reference group. Matching, GEE, regression and the versus-reference lines of tables and figures compare each group with it.

Also set the patient ID, so both eyes of one patient are clustered correctly.

2 Variables & Types

Put each variable into Baseline, Preop / Intraop or Outcome, and give it a type: continuous · ordinal · dichotomous · categorical.

The type decides the summary and the test. When a patient has two analyzed eyes, the P value of a per-eye row comes from a patient-clustered GEE of that type.

Say how your file writes visual acuity: decimal, logMAR, Snellen or ETDRS letters. CF, HM, LP and NLP become missing.

Acuity is analyzed in logMAR by default and shown as “20/20 (0.00 ± 0.05 logMAR)”: the Snellen geometric mean with the logMAR mean ± SD.

Steps 3–4 · Filters and matching

3 Filters & Subsets

Add inclusion rules — age, follow-up, first treatment, cylinder limits, and so on. A blank cell passes a rule (add a notna rule to drop blanks), and rules use the values as recorded in your file.

Row counts update live, per group, so you can see exactly how many eyes each rule removes before running anything.

Each enabled subset (e.g. PRK and LASIK) gets its own full analysis. Rows outside every subset are left out. The Workbooks setting can add, or give only, one pooled workbook.

4 Matching

Matching against the reference group: nearest neighbor, exact, coarsened exact or subclassification.

Choose the covariates and, for nearest neighbor, the ratio and the caliper. Default: 1:1, caliper 0.2 SD of the logit of the propensity score (Austin 2011); the probability scale of 1.0.0 stays selectable under Distance.

For nearest neighbor a live preview shows how many eyes match before you run.

Balance is reported on each pairing's PSM sheet and in its love plot, which name the scale used.

Steps 5–6 · Regression and GEE

How to handle groups that still differ at baseline

GEE is how the regression is fitted, not an alternative to it. The model is outcome ~ group + covariates; GEE additionally accounts for both eyes coming from the same patient. Every GEE test and interval uses the Mancl-DeRouen small-sample correction with a t reference.

- **Adjust for what is unbalanced** — Put the baseline variables that still differ into the regression covariates. The Regression sheet also gives the crude estimate fitted on the same eyes.
- **Match first, then adjust** — Set Run on to Matched cohort (or Both) on the Regression tab. Matched models use an independence working correlation, clustered by patient.
- **Choose covariates by balance, not by P** — Add a covariate because it is still imbalanced (SMD above ~0.1), never because it changes the result.
- **Small groups carry a caution** — A clustered model needs at least 6 eyes and 3 patients, otherwise it reports Not estimable. A group with 2-9 patients gets a caution under Check before using the results.
- **Settings on the GEE tab** — Run GEE on the matched cohort (default), the unmatched one or both. Summary-table tests: auto (clustered) or classical. P value style: journal (.27) or classic (0.266).

Step 7 · Special outputs and figures

Map the Column roles at the bottom of the tab first. Nothing is guessed from column names.

The composite

EVERY group on one chart, readable up to about five. One overall test plus, with three or more groups, each group against the reference (Holm-adjusted), clustered by patient when a patient ID is set. Panel sets: Full (ten panels), Refractive (six), Cataract (four). Stability joins the Full set as an eleventh panel.

Journal style

The default: white, no frames, section bands or shaded wedges. Each figure's title, method notes and abbreviations go to <figure>_legend.txt beside it. Figure style annotated keeps the 1.0.0 look, with that text inside the image.

Width and files

TIFF (300 dpi by default, up to 1200), PNG, JPEG, PDF or EPS. A figure width of 8.76, 12.69 or 17.77 cm (the journal columns) keeps text at 6 pt or more: two panels per row (one at 8.76 cm), and pages _p1, _p2 when taller than 23 cm.

Semicircles, export tabs

Semicircle plots of TIA, SIA and DV at plus axes, kept for familiarity. Optional tabs mEYEstro_<group> and AstigMATIC_<group> (matched pairing k: mEYEstro_M<k>_<group>) for cross-checking those tools.

Every panel is also saved as its own file, and the composites are previewed on the Run & Results tab.

Step 7 · Astigmatism (JCRS 2025)

Plus-cylinder vectors, whatever the cylinder form of each row: with-the-rule on the left, against-the-rule on the right

Double-angle plots

PRE (preoperative astigmatism), SIA (postoperative − preoperative) and PE, the prediction error (postoperative − target); POST joins when a target is not plano. Per cohort: centroid, total SD, magnitude mean ± SD, median and range.

Two 95% regions

Of the eyes: the depth-based convex polygon. Of the centroid: the bootstrap convex polygon of 999 bootstrap centroids. Both are built as in the WHWK functions, the journal's reference tool; the normal and Hotelling ellipses stay as options.

The tests

Each centroid against zero (bootstrap projection-distance test; P = .001 is its floor). Between cohorts: centroid, precision and mean magnitude by robust tests that resample patients, and the share of eyes ≤ 0.25 … ≤ 2.00 D. Holm-adjusted.

Sheets, WHWK input

“Astigmatism (JCRS 2025)”, also written as a CSV, and “Astigmatism vectors” (one row per eye and vector). A WHWK input folder beside the figures holds <vector>_vector.txt and <vector>_magnitude.txt for the WHWK R functions (osf.io/xhe8u).

The correction-index polar plots are gone (its histogram stays in the composite). Hull peeling is no longer offered.

Step 8 · Run, preview, save

Preview first, or save at once

Set the output folder in the left sidebar. ▶ RUN & SAVE writes the results straight there.

Tick Preview only to keep them inside the app until your next run or until you close it. Then press Save ALL to output folder (figures, love plots and CSV files too), or Save this one for a single workbook.

The Run & Results tab first lists how the numbers will be computed (Before you run); the Manifest sheet records the same.

Progress and a full log are shown while R runs. Cancel stops it.

What you get

- Excel workbooks — summary, Publication Table, matching, GEE, regression and the two astigmatism sheets, with a Contents and a Manifest sheet
- A matched-only workbook with every pairing
- A love plot per matched pairing: LovePlot_<prefix>_<run>_<pairing>_<time stamp>.tiff, with its legend file
- A figure folder per run, with the legend files and the WHWK input folder. The combined figures are previewed in the app
- The astigmatism sheet as CSV, and the analyzed rows as CSV (optional)

Presets — one per study

Save every setting of a project and reload it in one click

A preset stores everything: groups, variable typing, filters, subsets, matching, regression and figure choices.

- **Save** — Name the study and click save. It is stored inside the app on your own computer.
- **Reload** — Load the Excel file, then click the preset to restore the entire configuration. A preset never loads the data file itself.
- **Share** — Click Download config JSON and send it to a colleague. They click Load config JSON and get your exact setup. The data file is not inside, so they load their own copy of it.

Seven worked ‘Example – …’ presets are included. Load Example Data\Example_Refractive_Dataset.xlsx (simulated cohorts Reference, Cohort A and Cohort B; its README sheet gives the model), click one, tick Preview only and press RUN.

If something goes wrong

“Windows protected your PC”	Click More info, then Run anyway. To avoid it, tick Unblock in the ZIP’s Properties first.
“Windows security blocked a component”	Smart App Control or an antivirus blocked the runtime. Ask IT to allow this folder.
“Python runtime is missing”	The folder was copied incompletely, or is still inside a ZIP. Copy the whole folder again.
“Cannot write to its own folder”	Move the whole folder to Documents or C:\StandardEyes, not C:\Program Files.
The browser shows nothing	Wait a few seconds, then reload the address printed in the console window.
“R is not set up yet”	Click ⚙ R setup at the top right and point it at Rscript.exe.
A red “out of date” bar	Close the console window, start the app again, then press Ctrl+F5.
The analysis fails immediately	Check that R is version 4.5.x and the VA notation of your file. The log says why.
Am I in the right app?	The header shows StandardEyes and a version such as v1.1.0 (build 2026-10-04-se.10).

Everything runs on your own computer.

No data leaves the machine, no internet connection is needed, and no packages are ever installed.

Newer versions: standardeyes.github.io

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